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26 April 1973

Superseding
MIL-C-55086(SigC)
26 October 1959

MILITARY SPECIFICATION

CAMERA SET, STILL PICTURE, VIEW, 4 BY 5 INCH KS-17( )

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for a 4 x 5 inch, commercial type (see 6.4.1) metal, view camera with ancillary equipment and accessories, designated Camera Set, Still Picture, View, 4 By 5 Inch KS-17( ). (See 3.1 and 6.3.)

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Materials-handling, Wood (General Construction Requirements)
QQ-S-781	Strapping, Steel, Flat And Seals
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated-plywood
PPP-B-621	Box, Wood, Nailed And Lock Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-wall
PPP-F-320	Fiberboard, Corrugated And Solid, Sheet Stock (Container Grade) And Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-S-760	Strapping, Nonmetallic, (And Connectors)

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FEDERAL

PPP-T-76	Tape, Pressure-sensitive Adhesive Paper, Water Resistant, (For Carton Sealing)
PPP-T-97	Tape, Pressure-sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-675	Coating Of Glass Optical Elements (Antireflection)
MIL-H-3526	Holders, Photographic Sheet Film, Double, Lock Rib
MIL-F-9329	Filter, Light, Photographic, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-147	Palletized And Containerized Unit Loads 40 Inch X 48 Inch Pallets, Skids, Runners, Or Pallet Type Base
MIL-STD-150	Photographic Lenses
MIL-STD-252	Classification Of Visual And Mechanical Defects For Equipment, Electronic, Wired, And Other Devices
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-1278	Filters, Light, Photographic
MS-21329	Holder, Photographic Sheet Film, Double, Lock Rib

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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NATIONAL BUREAU OF STANDARDS

Handbook H-28 - Screw Thread Standard for Federal Services (Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 Description.- Camera Set, Still Picture, View, 4 by 5 Inch KS-17( ) is a metal, nominal 4 by 5 inch negative size, commercial-type view camera, a set of lensboard-mounted lenses in between-the-lens shutters, film holding devices, a tripod, and ancillary accessory items.

3.1.1 Components.- The camera set shall consist of the following:

a. A metal, commercial type, 4 by 5 inch, monorail, view camera with a lensboard-mounted, nominal 6 inch focal length, f/4.5 or faster lens in a between-the-lens shutter, and front and rear lens caps.

b. A lensboard-mounted, nominal 3-1/2 inch focal length, f/6.3 or faster, wide-angle lens in a between-the-lens shutter, and front and rear lens caps.

c. A lensboard-mounted, nominal 8-1/2 inch focal length, f/8 or faster lens in a between-the-lens shutter, and front and rear lens caps.

d. Six 4x5 inch sheet film holders. (See 6.2.)

e. A 4x5 inch film pack adapter.

f. Cable release.

g. Lens shades.

h. Filters.

i. Focusing cloth.

j. Camera carrying case.

k. Tripod with carrying case.

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3.2 First article.- When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.3 Parts, materials, and processes; general.- Parts, materials, and processes shall be such as to enable the equipment to meet its performance and other requirements as specified herein.

3.3.1 Tropicalization.-

3.3.1.1 Treatment of materials.- Fungus-inert materials shall be used where practicable. These materials shall be in accordance with requirement 4 of MIL-STD-454. When materials are required to be treated, they shall be cleaned in accordance with requirement 9 of MIL-STD-454 and then shall be treated in accordance with 3.3.1.2 through 3.3.1.5 herein.

3.3.1.2 Treating compounds.- Treating compounds containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating compound is compatible with the material or surface to be treated. Selection of the treating compound shall be such that there shall be no increase in flammability of the treated material.

3.3.1.3 Toxicity.- Treating compounds shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.3.1.4 Flexibility.- Treatment shall not affect the flexibility of the treated material to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.3.1.5 Statement of treatment.- The statement of treatment shall indicate whether or not nutrient materials will be used. If nutrient materials will be used, the statement shall also describe in detail the materials to be treated and the treating compounds and processes that the contractor proposes to use. (See 6.2(f).)

3.3.1.6 Fungus.- The equipment, including ancillary items and accessories, shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.9. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

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3.3.2 Threaded parts.- Screws, nuts, bolts, and other threaded parts shall be in accordance with Handbook H-28; the Unified Thread Form and Series shall be class 2A or 2B fit, and the American National Thread Series shall be class 2 fit. Wherever practicable, screw threads shall be commerical standard sizes. Threads in soft material shall be of the American National Coarse-Thread series; threads in other material shall be of either the American National Coarse-Thread or Fine-Thread series.

3.3.3 Length and engagement of threads.- Thread engagement shall be at least three complete threads. Threaded parts shall be properly secured by means of lock washers, stop nuts, or other suitable means.

3.4 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454.

3.5 Rigidity and alignment.- The design and construction of the camera and tripod shall be of sufficient rigidity and strength to maintain all parts in the alignment necessary to meet all requirements and tests specified herein.

3.6 Construction.-

3.6.1 Camera body.- The camera body shall consist of a front carriage, a bellows, and a rear carriage, mounted on a monorail bed fitted with an adjustable tripod mount. The basic structural elements of the camera shall be made of light-weight metal. Focussing and composing shall be by inspection of the image on a ground glass focussing screen. The camera design shall permit both, photography at a 1:1 image-object size ratio when the 8-1/2 inch focal length lens is used and maximum swings, tilts, and shifts are employed, and photography at infinity focus using the 3-1/2 inch focal length lens.

3.6.1.1 Light-tightness.- The camera shall be entirely light-tight to visible, ultra-violet, and infra-red radiation when tested in accordance with 4.13 herein.

3.6.1.2 Controls.- Controls and fastening devices shall be large enough to permit easy manipulation and shall be readily accessible. The setting, positioning, or adjustment of the controls and fastening devices shall not be inadvertently affected during normal use of the camera. The fastening devices shall operate easily and hold the movable or separable parts firmly to the mounting structure under all operating conditions.

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Fastening devices shall be of the captive type, remaining attached to the camera even in an unlocked condition.

3.6.1.3 Dimensions and weight.- The overall dimensions of the camera (exclusive of the bed) when in the closed position shall not exceed 12 inches in height, 8-1/2 inches in width, and 7 inches in depth. The weight of the unloaded camera (including the bed and tripod mount, but excluding the lensboard-mounted lens-shutter assembly) shall not exceed 9-1/2 pounds.

3.6.1.4 Picture area.- The camera shall provide an actual film image area not less than that permitted by the sheet film holder specified herein.

3.6.1.5 Rear carriage.- The rear carriage shall be of rigid construction and shall be movable along the bed by means of a rack and pinion drive or other means which shall provide similar positive control of movement. A means shall be provided for locking the rear carriage in any location on its length of travel along the bed. When locked, the carriage shall remain firmly in place. The rear carriage shall incorporate the rear standard.

3.6.1.5.1 Rear standard.- The rear standard shall furnish rigid support for the camera back and shall be suitably attached to the rear carriage. The standard shall be provided with a fixed spirit level(s) mounted in such manner as to facilitate horizontal and vertical leveling of the standard. The standard shall be provided with clips or other suitable means for attachment of the focussing cloth specified herein.

3.6.1.5.1.1 Horizontal swing.- The rear standard shall permit swinging the camera back at least 12 degrees in either direction on a vertical axis through the center of the film plane. A detent or other suitable positive locating facility shall be provided to orient the swing in the neutral position.

3.6.1.5.1.2 Vertical tilt.- The rear standard shall permit tilting the camera back at least 26 degrees in either direction on a horizontal axis through the center of the film plane. A detent or other suitable positive locating facility shall be provided to orient the tilt in the neutral position.

3.6.1.5.1.3 Lateral shift.- The rear standard shall permit lateral shifting of the camera back at least 5/8 inch to either side of the camera's longitudinal center line. A detent or other suitable positive locating facility shall be provided to orient the shift in the neutral position.

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3.6.1.5.2 Rear movement locks.- Each facility provided for movement of the rear carriage, rear standard, and camera back shall be provided with its individual positive means of locking each movement in its set position. Each means provided shall be readily accessible and capable of operation without disturbance of any other camera setting.

3.6.1.6 Camera back.- The camera back shall be of the revolving type, rotatable at least 90 degrees each, clockwise and counter-clockwise, about the center point of the image plane without removal from the rear standard. The back, which shall incorporate a viewing-focussing panel, shall accommoderate 4 by 5 inch sheet film holders and film pack adapters as specified herein. The back shall permit the utilization of the full picture area of the film as specified herein regardless of the back's orientation. Detents or other suitable locating facilities shall be provided to orient the back in the vertical and horizontal format positions. A positive means shall be provided to lock the back at any set degree of rotation.

3.6.1.6.1 Viewing-focussing panel.- The viewing-focussing panel shall be non-detachable. The panel shall consist of a fine-ground glass focussing screen with a fresnel type condenser field lens in a metal frame. The screen shall be scored or otherwise permanently marked at 1/2 inch intervals with vertical and horizontal lines parallel to the long and short sides of the screen. The ground glass and field lens shall be fastened to the focussing panel in a manner which shall permit their replacement without the use of special tools. The means provided for movement of the panel for the easy insertion and withdrawal of the film holding devices specified herein shall supply sufficient force to hold the screen and the film holding device in the plane necessary for sharply focussed images.

3.6.1.7 Bellows.- The bellows shall be a light-tight, collapsible enclosure connecting the front and rear standards. It shall be of suitable flexible but tough material to permit at least 5,000 cycles of complete compression and extension without cracking, tearing, developing pinholes, or suffering any other loss of light-tightness. Although it shall be firmly attached to the front and rear standards, the method of attachment shall permit its ready replacement when necessary. The bellows' design, construction, and method of attachment to the standards shall not result in any interference with the image under any condition of the camera's movement settings.

3.6.1.7.1 Bellows size.- The dimensions of the bellows shall be suitable to permit the combined maximum movements of the front and rear

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standards' swings, tilts, and shifts in conjunction with maximum lens rise for photography at a 1:1 image-object size ratio using the 8-1/2 inch focal length lens. The combined bellows and standards dimensions shall permit the camera to be focussed for "infinity" distance when the 3-1/2 inch focal length lens is used.

3.6.1.8 Front carriage.- The front carriage shall be of rigid construction and shall be movable along the bed by means of a rack and pinion drive or other means which shall provide similar positive control of movement. A means shall be provided for locking the front carriage in any location along its length of travel on the bed. When locked, the carriage shall remain firmly in place. The front carriage shall incorporate the front standard.

3.6.1.8.1 Front standard.- The front standard shall furnish rigid support for the lensboard and shall be suitably attached to the front carriage. The standard shall be provided with a readily accessible, positive means for locking the lensboard in place.

3.6.1.8.1.1 Horizontal swing.- The front standard shall permit swinging the lensboard at least 12 degrees in either direction on a vertical axis through the rear nodal point of each of the lenses supplied. A detent or other suitable positive locating facility shall be provided to orient the swing in the neutral position.

3.6.1.8.1.2 Vertical tilt.- The front standard shall permit tilting the lensboard at least 26 degrees in either direction on a horizontal axis through the rear nodal point of each of the lenses supplied. A detent or other suitable positive locating facility shall be provided to orient the tilt in the neutral position.

3.6.1.8.1.3 Lateral shift.- The front standard shall permit lateral shifting the lensboard at least 5/8 inch to either side of the camera's longitudinal center line. A detent or other suitable positive locating facility shall be provided to orient the shift in the neutral position.

3.6.1.8.1.4 Rising-falling front.- The front standard shall have provision for moving the lensboard-mounted lens-shutter assembly at least three inches upward and one inch downward from its neutral position. The neutral position, which shall be located by means of a detent, markings, or other indicating facility, shall be on the horizontal axis extending through the lens nodal points and the center of the film plane. Movement of the

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rising-falling front shall be by rack and pinion or other positive means of control. A limit stop shall be provided to prevent excessive dislocation of the lensboard.

3.6.1.8.2 Front movement locks.- Each facility provided for movement of the front carriage, front standard, and lensboard shall be provided with its individual positive means of locking each movement in its set position. Each means provided shall be readily accessible and capable of operation without disturbance of any other camera setting.

3.6.1.9 Bed.- The camera bed, which shall be of the monorail type, shall serve as a rigid support for the front and rear carriages. The bed shall provide for individual forward and backward movement of each, the front and the rear carriages. Movement of the carriages shall be independent of the tripod mount. The fitting of the bed to the appropriate portions of the carriages shall not permit rotation of the carriages about the bed's longitudinal axis more than one degree in each direction. If a rack and pinion method of carriage movement is used, the rack shall be securely fixed within the bed.

3.6.1.9.1 Bed length.- The bed length shall be at least long enough to permit photography at a 1:1 image-object size ratio when the 8-1/2 inch focal length lens is used. The construction of the bed shall not permit extension of the bellows to an extent which would place an undue strain on the bellows or its points of attachment to the front and rear standards. The bed shall be provided with a positive stop at each end to prevent inadvertent removal of the carriages from the bed.

3.6.1.10 Tripod mount.- A tripod mount incorporating a tripod socket suitable for accepting a commercial standard 1/4-20 tripod screw shall be provided. The tripod mount shall be movable along the camera bed to permit placement of the mount beneath the center of gravity of the camera set at any usable extension. A means shall be provided for securely locking the mount at any location along its length of travel on the bed. The fitting of the unlocked mount to the bed shall not permit rotation of the bed about its longitudinal axis more than one degree in each direction.

3.7 Lens-shutter-lensboard assemblies.-

3.7.1 Lenses.- (See 6.4.2.)

3.7.1.1 Type.- Lenses shall be type III, anastigmat, in accordance with MIL-STD-150.

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3.7.1.2 Focal length and aperture.- The equivalent focal lengths and maximum relative apertures of the lenses shall be in accordance with table I. The acceptable range of marked focal lengths shall be as follows: nominal 6 inch focal length - 150 to 165 millimeters (mm); nominal 3-1/2 inch focal length - 90mm; nominal 8-1/2 inch focal length - 210 to 225mm. The percentage difference between actual focal length versus marked focal length shall be in accordance with MIL-STD-150.

Table I.-

Nominal Equivalent Focal Length	Maximum Relative Aperture
3-1/2 inch	f/8 or faster
6 inch	f/5.6 or faster
8-1/2 inch	f/5.6 or faster

3.7.1.3 Relative illumination.- The relative illumination of the lenses shall be at least in accordance with tables II, III, and IV when measured in accordance with 4.17 herein.

Table II.-

3-1/2 Inch Lens

Angle (in degrees)	0	8	16	24	32	39	45	50
Resolving power (in lines per millimeter) at working maximum relative aperture	60	38	24	17	17	34	15	10
% Relative illumination at working maximum relative aperture	100	99	98	97	90	75	50	33
% Relative illumination at f/22	100	99	98	97	95	90	67	50

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Table III.-

6 Inch Lens

Angle (in degrees)	0	5	10	15	20	25	30	35
Resolving power (in lines per millimeter) at maximum relative aperture	35	35	30	20	18	15	13	15
% Relative illumination at maximum relative aperture	100	99	98	97	93	88	67	50
% Relative illumination at f/22	100	99	98	97	95	93	82	75

Table IV.-

8-1/2 Inch Lens

Angle (in degrees)	0	5	10	15	20	25
Resolving power (in lines per millimeter) at maximum relative aperture	53	47	17	9	11	14
% Relative illumination at maximum relative aperture	100	99	98	97	94	50
% Relative illumination at f/22	100	99	99	98	97	95

3.7.1.4 Resolving power.- The resolving power of the lenses shall be at least in accordance with tables II, III, and IV when measured in accordance with 4.12 herein.

3.7.1.5 Diaphragm.- The lens assembly shall incorporate a continuously variable iris diaphragm. The operation of the diaphragm shall be accomplished by means of a manual control which shall reduce the aperture

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when the control is rotated counter-clockwise from a viewpoint in front of the lens. A limit stop shall be provided to prevent the closing of the diaphragm beyond the smallest marked aperture. The diaphragm control shall be click-stopped at each full f/stop marking falling between the largest and smallest values.

3.7.1.5.1 Aperture markings.- Aperture calibration shall be marked in the f/system and shall conform to MIL-STD-150. The smallest marked aperture shall be at least f/32. Fractional stop values, if used, shall be in accordance with MIL-STD-150.

3.7.1.6 Reflection reducing coating.- Each air-glass surface shall be reflection-reducing coated in accordance with MIL-C-675.

3.7.1.7 Beauty defects.- Beauty defects, as defined in MIL-STD-150, while undesirable, are acceptable to the extent that their presence shall permit the lenses to meet the requirement of 3.7.1.4 and test of 4.12.

3.7.1.8 Lens caps.- Two snug-fitting, velvet-lined, metal, lens caps shall be furnished with each lens. One cap shall be designed to protect the front element, and the other to protect the rear element of the lens.

3.7.2 Shutters.- Shutters shall operate at the effective exposure times within the tolerances specified herein. The shutters shall have built-in synchronization for zero delay and 20 millisecond delay flash lamps. Shutter operating controls shall include an exposure time setting control operating against an index marked with exposure times, a shutter cocking lever, a shutter actuating lever, and a flash synchronization delay selection control. The shutter shall be provided with a standard commercial cable release socket.

3.7.2.1 Shutter speed markings.- Shutter speed markings shall be "T" for time exposure, "B" for bulb exposure, and the denominator representing fractional second exposure times in either of the following standard commercial sequence systems: 1/1, 1/2, 1/4, 1/8, 1/15, 1/30, 1/60, 1/125, etc., or 1/1, 1/2, 1/5, 1/10, 1/25, 1/50, 1/100, etc.

3.7.2.2 3-1/2 inch lens shutter.- The range of marked exposure times shall be in accordance with 3.7.2.1, with the shortest exposure time being not longer than 1/400 (or 1/500) second. The effective exposure times, when measured at an opening with an area equal to that of the working

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maximum lens aperture, shall not depart from the marked exposure values by more than 20 percent for "instantaneous" exposures of 1/125 (or 1/100) second and longer, nor more than 30 percent for exposures shorter than 1/125 (or 1/100) second, at any temperature from plus 35 degrees Fahrenheit (F) to plus 120 degrees F. At temperatures from zero degrees F. to plus 35 degrees F. the effective exposure times shall not depart from the marked exposure values by more than 40 and 50 percent, respectively.

3.7.2.3 6 inch and 8-1/2 lens shutters.- The range of marked exposure times shall be in accordance with 3.7.2.1, with the shortest time being not longer than 1/250 (or 1/200) second. The effective exposure times, when measured at an opening with an area equal to that of the maximum lens aperture, shall not depart from the marked exposure values by more than 20 percent for "instantaneous" exposures of 1/60 (or 1/50) second and longer, nor more than 30 percent for exposures shorter than 1/60 (or 1/50) second, at any temperature from plus 35 degrees F. to plus 120 degrees F. At temperatures from zero degrees F. to plus 35 degrees F. the effective exposure times shall not depart from the marked exposure values by more than 40 and 50 percent, respectively.

3.7.2.4 Synchronizing mechanisms.- The synchronizing mechanism shall consist of built-in electrical contacts capable of accurately synchronizing 20 millisecond delay flash lamps and zero delay electronic flash units with the wide open position for any instantaneous shutter speed. Contact for the zero millisecond setting shall occur at the time the shutter reaches its maximum opening plus or minus one millisecond. Contact for the 20 millisecond setting shall occur at 17 plus or minus two milliseconds before the shutter reaches its maximum opening.

3.7.2.4.1 Flash cable connector.- Each shutter shall be provided with a commercial standard "PC" flash cable connector.

3.7.2.5 Cable release socket.- Each shutter shall be provided with a socket for attachment of a commercial, screw-in type of cable release.

3.7.2.6 Lensboard.- Each lensboard shall be constructed of a rigid, light-weight metal and shall fit the front standard. The lensboard shall be readily attachable to and detachable from the standard without the use of tools. The lensboard shall mount the lens-shutter assembly in such manner as to furnish suitable alignment of the lens with the film plane when all movements of the front and rear standards are in their neutral operating positions.

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3.8 Lens shade.- A suitably dimensioned, light-weight metal lens shade shall be supplied for each, the 6 inch, and the 8-1/2 inch focal length lens. The shades shall be of two piece construction, consisting of a screw-in or slip-on adapter and a screw-in hood having a black, matte finished, interior surface. The adapter ring shall accommodate drop-in filter discs which shall be held in place by the hood. The adapter ring shall be capable of accepting the hood when a filter is not used. The lens-mounted shade-filter assembly shall not cause optical or physical interference with the proper operation of the camera.

3.9 Filters.- A set of filters consisting of one each, round, metal-banded, drop-in, type II or type III, grade B filter conforming to MIL-STD-1278 and in accordance with MIL-F-9329 shall be supplied in the size(s) necessary to fit the 6 inch and 8-1/2 inch focal length lenses specified herein. There shall be no size duplication of any given color. Each set shall consist of a B-2-8, a B-2-15, and a Z-.60-4x filter, each packaged in its own transparent, rigid container or within its individual compartment of a case designed to contain the entire set(s). Filters for the 3-1/2 inch focal length lens shall conform to all of the foregoing except that they shall be of the screw-in construction, fitted to the integral lens shade of the lens, and that the size duplication restriction is not applicable.

3.10 Cable release.- A metal - or vinyl-sheathed, flexible cable release shall be supplied. The cable release tip and the length of "throw" shall be commensurate with the design of the shutters supplied. The cable release shall be a nominal 12 inches long.

3.11 Focussing cloth.- A black, closely-woven, soft, focussing cloth having bound or hemmed edges shall be supplied. The cloth shall be at least a nominal 40 by 48 inches in size.

3.12 Holder, sheet film.- Unless otherwise specified, 6 sheet film holders with slides, Size I, 4 inch by 5 inch, in accordance with MIL-H-3526 and tables I and II of Standard MS21329 shall be supplied. (See 6.2.) The holder shall accommodate two sheets of nominal 4 inch by 5 inch sheet film, one on each side of the holder. The holder shall be light-tight, suitable for use with infra-red sensitive film, and so constructed that film is held in its proper position in the focal plane during its exposure. The design of the dark slide shall provide for the visual and tactile identification of unexposed and exposed film contained within the holder. Provision shall be made to prevent inadvertent removal of the dark slide. Dark slides shall be opaque to infra-red radiation.

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3.13 Film pack adapter.- A film pack adapter accommodating commercially available standard 4 inch by 5 inch film packs shall be supplied. The adapter's construction shall be such as to position each successive sheet of film in the focal plane. The film pack adapter shall be supplied with a dark slide suitable for its intended purpose and shall incorporate a facility to prevent inadvertent removal of the slide.

3.14 Tripod.- An anodized aluminum alloy, elevator tripod having a 360 degree panning, and forward - and backward-tilting head shall be supplied. The overall length of the collapsed tripod shall not exceed 30 ± 1/2 inches. The maximum height when properly set up for use shall be at least a nominal 6 feet. The head shall permit the camera to be tilted at least 90 degrees downward, and a nominal 50 degrees upward from the horizontal. A positive locking means shall be provided for each, the panning and the tilting movement. The camera-mounting screw shall be a commercial standard 1/4-20 thread and shall have a facility for adjustment of the length of camera engagement.

3.14.1 Legs.- Each tripod leg shall consist of three sections - one fixed; two adjustable. The sections comprising each leg shall not be separable each from the other in normal operation. The ground-touching portion of each leg shall be fitted with a replaceable type of non-slipping, non-marring facility. The means provided for locking the legs at the set degree of extension shall be capable of easy operation via a heavy-gloved hand.

3.14.2 Elevator column.- The elevator column shall be capable of at least a 12 inch rise. Movement of the column shall be accomplished by means of a hand-operated gear drive. Means shall be provided to lock the column at any set point of extension. A limit stop shall be provided to prevent the inadvertent removal of the column from its supporting structure.

3.14.3 Tripod case.- A fungus-resistant canvas carrying case designed to house and protect the tripod shall be provided. The case shall be fitted with both a carrying handle located above the approximate center of gravity of the tripod, and a nominal 1 inch wide, adjustable length, fungus-resistant, web, shoulder carrying strap.

3.15 Carrying case.- A suitably fitted, sturdy, light-weight metal, luggage type carrying case with handle, designed to house and protect all parts of the camera and accessory items of this specification except the tripod and the tripod carrying case, shall be provided. (See 3.1.1.)

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The exterior and interior surfaces of the case shall be as smooth as possible. Cases with projecting rivets will not be acceptable. The case shall be provided with metal loops through which the web shoulder strap specified herein may be threaded.

3.15.1 Closure.- The carrying case shall have a piano-type hinge, and the closure shall effectively seal the case against penetration of wind-driven rain. Drawbolt type fasteners shall be provided to keep the case closed. A means shall be provided to hold the lid open at an approximately right angle to the bottom of the case.

3.15.2 Interior fittings.- Provision shall be made to store and protect all parts of the equipment, exclusive of the tripod and the tripod carrying case. Necessary pads, partitions, clamps, and hold-down straps shall be provided to hold all parts of the equipment securely in position. The camera storage facility shall be so located that no portion of the camera shall make contact with any exterior wall of the case. A laminate-protected list of case contents, including the item name, quantity, and federal stock number of each item shall be fixed to the inner surface of the case lid. The equipment shall be so arranged that the center of gravity of the loaded case shall be approximately beneath the carrying handle.

3.15.3 Carrying devices.- The carrying case shall be provided with a hand-fitting, rigid, luggage-type handle. The handle shall be capable of being folded down when not in use. A nominal 1 inch wide, adjustable length, fungus resistant, web, shoulder carrying strap designed to fit through the loops provided on the carrying case shall be provided.

3.16 Service conditions.- The equipment shall meet the following service conditions:

3.16.1 Temperature.-

3.16.1.1 Low.- During testing as specified in 4.16.1 the equipment shall meet the following requirement:

Paragraph	Measurement	Performance
4.16.1	Shutter speed at 35°F.	3.7.2.2; 3.7.2.3
4.14	Operation	Full specification requirements

After testing, the equipment shall meet full specification performance for these measurements.

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3.16.1.2 High.- During testing as specified in 4.16.2 the equipment shall meet the following requirement:

Paragraph	Measurement	Performance
4.16.2	Shutter speed at 120°F.	3.7.2.2; 3.7.2.3
4.14	Operation	Full specification requirements

After testing, the equipment shall meet full specification performance for these measurements.

3.16.2 Humidity (moisture-resistance).- After testing as specified in 4.10 the equipment shall meet full specification performance for the following measurement:

Paragraph	Measurement
4.14	Operation

3.16.3 Altitude (elevation).- After testing as specified in 4.6 the equipment shall meet full specification performance for the following measurement:

Paragraph	Measurement
4.14	Operation

3.17 Marking.-

3.17.1 Lens.- The location of lens markings such as maximum relative aperture, nominal focal length, serial number, trade name, cell markings, etc. shall be in accordance with MIL-STD-150 and shall conform to standard commercial practice as regards the format of the data presented.

3.17.2 Shutter.- Each shutter shall be serial numbered in such location as to be readily visible when the lensboard-mounted lens-shutter assembly is installed in the camera.

3.17.3 Camera set identification marking.- The exterior top surface of the camera set carrying case shall bear a permanently fixed nameplate conforming to MIL-STD-130.

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3.18 Workmanship.- The equipment shall be manufactured and assembled in accordance with the requirements of MIL-STD-454 and the applicable portions of the following indicated paragraphs herein:

3.3.2 Threaded parts

3.3.3 Length and engagement of threads

3.6.1.2 Controls

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections.- The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.3.) Does not include preparation for delivery.

b. Inspections covered by subsidiary documents. (See 4.4.)

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5.)

(2) Quality conformance inspection of preparation for delivery. (See 4.19.)

4.3 First article.- Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

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4.3.1 First article units.- The contractor shall furnish two (2) first article units of the complete Camera Set, Still Picture, View, 4 by 5 Inch KS-17( ).

4.3.2 First article inspection.- The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, and the inspections specified for Group A, Group B, and Group C. (See tables V, VII, and VIII respectively.) The inspection shall be performed in the following order: Group A, Group B, and Group C. After completion of all Group C tests, conforming units shall be reinspected and pass all Group A inspection.

4.4 Inspections covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where Required
Aperture marking	3.7.1.5.1
Reflection reducing coating	3.7.1.6
Beauty defects	3.7.1.7
Sheet film holders	3.12
Filters	3.9
Marking	3.17.1, 3.17.3

4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection.- Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in table V. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in table V.

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Table V.- Group A inspection

Inspection	Requirement Para	Test Para	AQL Major	AQL Minor
Visual and Mechanical	3.18	4.18	1%	4%
Camera body	3.6.1 - 3.6.1.4	4.18	1%	4%
Rear carriage	3.6.1.5 - 3.6.1.5.2	4.18	1%	4%
Camera back	3.6.1.6 - 3.6.1.6.1	4.18	1%	4%
Bellows	3.6.1.7 - 3.6.1.7.1	4.18	1%	4%
Front carriage	3.6.1.8 - 3.6.1.8.1.4	4.18	1%	4%
Bed	3.6.1.9 - 3.6.1.9.1	4.18	1%	4%
Lenses	3.7.1.1 - 3.7.1.5	4.18	1%	4%
Lenscaps	3.7.1.8	4.18	1%	4%
Cable release	3.10	4.18	1%	4%
Focusing cloth	3.11	4.18	1%	4%
Film pack adapter	3.13	4.18	1%	4%
Tripod	3.14 - 3.14.3	4.18	1%	4%
Carrying case	3.15 - 3.15.3	4.18	1%	4%
Operation	3.6.1.5 - 3.7.2.6	4.14	1%	*

* Defects in all characteristics other than visual and mechanical are considered major

4.5.1.1 Order of inspection within Group A.- Group A inspection shall be performed in an order satisfactory to the Government, except that the operational in 4.14 shall be last.

4.5.2 Group B inspection.- Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. This inspection shall conform to Table VI and to the special inspection levels of Table I of MIL-STD-105:

4.5.2.1 Group B sampling plans.- The Group B sampling plans, for the AQL's listed in table VI, shall be as follows:

AQL	Inspection level
4.0%	S-4
6.5%	S-4

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4.5.2.2 Order of inspection within Group B.- Group B inspection shall be performed in an order satisfactory to the Government.

Table VI.- Group B inspection

Inspection	Requirement Para	Test Para	AQL
Lens resolution	3.7.1.4	4.12	4.0%
Shutter speed	3.7.2.2; 3.7.2.3	4.15	4.0%
Circle of illumination	3.7.1.3	4.17	4.0%
Flash synchronization	3.7.2.4	4.8	4.0%
Light tightness	3.6.1.1	4.13	4.0%
Interchangeability	3.4	4.11	6.5%

4.5.3 Group C inspection.- Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table VII. Samples shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for Group C inspection.- One sample unit shall be selected at random for all inspections in Table VII from every 50 and fraction thereof produced. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.3.2 Order of inspection within Group C.- Group C inspection shall be performed in accordance with Table VII.

4.5.3.3 Group C failures.- Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group B and C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Groups B and C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Altitude (Elevation) Test.- The equipment shall be tested in accordance with Method 500, Procedure 1, of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.16.3.

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Table VII.- Group C inspection

Inspection	Requirement Paragraph	Test Paragraph
Subgroup 1		
Bellows Life	3.6.1.7	4.7
Subgroup 2		
Non-damaging (Note 1)		
Temperature		
Low	3.16.1.1	4.16.1
High	3.16.1.2	4.16.2
Altitude	3.16.3	4.6
Potentially damaging		
Humidity (moisture-resistance)	3.16.2	4.10
Fungus (Note 2)	3.3.1.6	4.9

NOTE 1: Other preproduction inspection may precede, follow, or be interspersed between, the Subgroup 2 tests.

NOTE 2: The equipment shall be thoroughly washed, cleaned, dried, and if necessary, refurbished before proceeding with subsequent inspection.

4.7 Bellow life.- The bellows shall be fully extended and completely compressed 5,000 times (1500 times @35°F; 1500 times @120°F and 2000 times at 70°F) to demonstrate compliance with 3.6.1.7. At the completion of the 5,000 cycles, the camera shall be subjected to the light-tightness test. (See 4.13.) Any light leakage shall be considered a failure.

4.8 Flash synchronization.- The shutters shall be tested to determine proper "M" and "X" synchronization. The test shall include measurement of delay time, shutter contacts duration, efficiency of the shutter contact, and contact insulation value. The test results shall demonstrate compliance with 3.7.2.4.

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4.9 Fungus test.- The equipment (including accessories) shall be tested in accordance with Method 508, Procedure 1, of MIL-STD-810. Spraying shall include parts and materials on and inside the camera, and the interior of the carrying case. The equipment shall remain in an open condition during the test. Operation is not required. (See 3.3.1.6.)

4.10 Humidity (moisture-resistance) test.- The equipment shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.16.2. Measurements shall be made after completion of the test.

4.11 Interchangeability.- The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.4. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, fasteners, slides, and mountings which receive mating parts of assemblies and major units; and location of mating parts on the assembly or major unit.

e. Size and form of special threads.

4.12 Lens resolution.- The lens shall be tested in accordance with Method II of MIL-STD-150. The film utilized shall be Panatomic X film and it shall be processed in D-76 developer to a gamma of 0.9 (see 3.7.1.4).

4.13 Light-tightness.- The camera shall be tested to determine that it is light-tight to visible, infrared, and ultra-violet light. The test shall be performed using infrared film loaded in the camera. The camera shall be mounted on the tripod with the bellows fully extended, and the dark slide withdrawn halfway. Each side top and bottom of the camera shall be exposed to a light source, ASA Lamp Code EBV Photo lamp in a metal

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reflector, at a distance of 6 ft, for a period of 5 minutes (a total of 30 minutes). The test film shall be processed, following the film manufacturer's instructions, and examined. There shall be no evidence of exposure on the processed film. (See 3.6.1.1.)

4.14 Operational test.- The camera set shall be given an operational test which includes, as a minimum:

a. Operation of all camera movements, e.g. rising-falling front, front and rear swings, tilts, and lateral movements.

b. Extension and closure of the bellows.

c. The insertion of film holders and film pack adapter.

d. The making of exposures.

e. Operation on the tripod.

4.15 Shutter test.- The shutter shall be operated ten times at each exposure time prior to the testing. The shutter shall then be evaluated for effective exposure at each setting at temperatures of 35°F, 70°F, and 120°F.

4.16 Temperature.-

4.16.1 Low temperature.- The equipment shall be tested in accordance with Method 502, Procedure 1, of MIL-STD-810. The storage temperature shall be -65°F and the operating temperature shall be +35°F. Prior to testing the equipment shall meet full specification for the measurements specified in 3.16.1.1.

4.16.2 High temperature.- The equipment shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. The operating temperature shall be 120°F. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.16.1.2.

4.17 Circle of illumination test.- The relative illumination of each lens shall be tested in accordance with Method 7, 8, or 9 of MIL-STD-150 to demonstrate compliance with 3.7.1.3.

4.18 Visual and mechanical.- Equipment shall be examined for the defects listed in MIL-STD-252 and Table VIII.

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Table VIII.- Classification of visual and mechanical defects

Component	Classification	Defect
Camera body	Major	Light leaks in body, bellows or fittings. Inoperable adjustment controls. Insufficient tension of film holder springs. Damaged tripod fittings.
	Minor	Sticking or binding adjustment controls.
Lenses	Major	Poor coating application. Shutter fails to trip. Shutter fails to remain in cocked position. Synchronizer adjustment inoperable. Shutter leaves do not close completely. Diaphragm closes down completely. Diaphragm sticks, binds or does not close concentrically.
	Minor	None
Film Holders	Major	Cracks, holes or light leaks in slides, septum or cross strip. Sticking or binding of slide in film holder.
	Minor	None
Focusing cloth	Major	Rips or tears.
	Minor	None
Tripod	Major	Horizontal and vertical locking device of head inoperable. Adjustable legs or elevator column cannot be extended or contracted.
	Minor	Binding of head movement. Binding of tripod legs.

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Table VIII.- Classification of visual and mechanical defects

Component	Classification	Defect
Carrying Case	Major	Inability of cover to close completely or lock.
	Minor	Doesn't open or close freely. Accessories do not fit properly. Hardware and clamping devices missing or loose. Cover does not fit snugly.

4.19 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of section 5.

4.20 Rough handling test (preparation for delivery).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract (see 6.2(d)) the following functional test shall be conducted to determine freedom from operational malfunction caused by the rough handling: Operational test, paragraph 4.14 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.2(c)).

5.1.1 Level A.-

5.1.1.1 Cleaning.- Camera Set, Still Picture, View, 4 by 5 inch; KS-17( ) shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying.- Camera Set, Still Picture View, 4 by 5 inch, KS-17( ), shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application.- None required.

5.1.1.4 Unit packaging.- Unit packaging shall be in accordance with the method prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature.- Each technical literature shall be packaged method IC-1.

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5.1.1.4.2 Tripod.- Each tripod shall be individually packaged method III as follows: Place the collapsed tripod within the camera carrying bag and secure cover with fasteners provided. Cushion each canvas carrying case on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, variety SW, class weather-resistant, grade W5c, designed to protect all projections and absorb the shock of impact in handling and transit. Place the cushioned item within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Close the box in accordance with the appendix of the box specification.

5.1.1.4.3 Camera Set, Still Picture, View 4 by 5 inch, KS-17( ).- Each camera set, still picture, view 4 by 5 inch, KS-17( ) with components and spare tissues shall be stored within the designated spaces in the carrying case. Fill all voids with paperboard conforming to PPP-P-291, type I, to prevent movement. Close cover of carrying case and secure with fasteners provided. Fabricate corner pads of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c. Secure the corner pads to the case with tape conforming to PPP-T-76.

5.1.1.4.4 Consolidated package.- Place the items, packaged as specified in 5.1.1.4.2 and 5.1.1.4.3, together within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Fill all void space with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c to prevent movement. Place the technical literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Closure shall be in accordance with the appendix of the box specification.

5.1.2 Level C.- Camera Set, Still Picture, View 4 by 5 inch, KS-17( ), shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing.- Packing shall be level A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container or its contents (see 6.2(c)).

5.2.1 Level A.-

5.2.1.1 Palletized load.- A quantity of Camera Set, Still Picture, View 4 by 5 inch, KS-17( ), packaged as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147 except that the pallet shall

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be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "banded" to the pallet by strapping.

5.2.1.2 Less than palletized load.- When quantities per destination are less than a pallet load, the items packaged as specified in 5.1, shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48x24 inches or more and the weight exceeds 200 pounds, or the container length and width is 48x24 inches or more and the weight exceeds 100 pounds, 3x4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, class B.

5.2.2 Level B.-

5.2.2.1 Palletized load.- A quantity of Camera Set, Still Picture, View 4 by 5 inch, KS-17( ), packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1.

5.2.2.2 Less than palletized load.- When quantities per destination are less than a pallet load, a quantity of Camera Set, Still Picture, View 4 by 5 inch, KS-17( ), packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48x24 inches or more and the weight exceeds 100 pounds, containers will be pallet mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or non-metallic strapping conforming to PPP-T-97, type IV and PPP-S-760, type II, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

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5.2.3 Level C.-

5.2.3.1 Palletized load.- A quantity of Camera Set, Still Picture, View, 4 by 5 inch, KS-17( ), packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1 except that the fiberboard caps shall be class domestic.

5.2.3.2 Less than palletized load.- When quantities per destination are less than a pallet load, a quantity of Camera Set, Still Picture, View 4 by 5 inch, KS-17( ), packaged as specified in 5.1, shall be packed as specified in 5.2.2.2 except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, class 1 and class domestic, respectively, and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The camera set is used for studio and general view photography where there is a need for correct representation of three-dimensional objects on a single plane without distortion of vertical and horizontal lines, and a control of depth of field beyond that which is a function of lens focal length, f/stop, and object distance.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Quantity of sheet film holders required for each camera set required. (See 3.12.)

c. Level of preservation and packaging and packing. (See section 5.)

d. When rough handling and functional tests are required.

e. First article (preproduction) inspection:

Two sample units of each item are generally required so that lengthy environmental tests can be completed on one sample unit while complete performance measurements can be made on the second sample unit. (See 3.2.)

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f. Submission of the statement of treatment referenced in 3.3.1.5, as soon as possible after award of contract. This statement shall be submitted to the contracting officer.

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: KS-17W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1.)

6.4 Definitions.- The following terminology as used herein is defined as follows:

6.4.1 Commercial-type-view camera - signifies a form of view camera whose front standard is capable of being adjusted to varying degrees of swing,tilt, shift, and rise/fall of front - as differentiated from "portrait-type view camera" whose front standard movement capability if any, is limited to a rising-falling front.

6.4.2 Optical terms.- The optical terminology used herein is in accordance with MIL-STD-150.

NOTE: Asterisks are not used in this revision to identify changes with regard to the previous issue due to the extensiveness of the changes.

Custodian:

Army-EL

Preparing Activity:

Army-EL

Project No. 6720-A022

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-C-55086A(EL) CAMERA SET, STILL PICTURE, VIEW, 4 BY 5 INCH KS-17( )

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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